# A biologist's guide to rapid and robust detection of GSK-J4

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**BACKGROUND**
Modulating epigenetic targets has shown therapeutic potential in various conditions, including inherited retinal disease. Of recent interest is the modulation of H3K27 trimethylation (H3K27me3). A seminal study by Miller *et al.* (2022) demonstrated that increasing H3K27me3 through continuous administration of GSK-J4 increases cone survival *ex vivo* in the *Pde6c*$^{cpfl1}$ mouse model of achromatopsia. Although GSK-J4 is a useful probe for modulating H3K27me3, developing it as a treatment requires improved physicochemical and pharmacokinetic characterisation, as well as a simple method to detect and quantify the drug. Here, we present a fast, robust HPLC method for quantifying GSK-J4 that is accessible to non-chemists.

**METHODS**
Various mobile phases, flow rates, and elution conditions were evaluated to develop the high-performance liquid chromatography (HPLC) method. The optimised method was then validated by assessing precision, accuracy, linearity, and specificity. Importantly, the method's compatibility with GSK-J4 extracted from treated 661W cone-like photoreceptor cells was also evaluated.

**RESULTS**
The optimised HPLC method met all acceptance criteria for precision, accuracy, linearity and specificity tests, highlighting its robustness. It employs simple mobile phases and isocratic conditions with rapid drug elution, enabling the detection of the drug in under 5 minutes. Notably, the method also allowed for successful quantification of GSK-J4 extracted from treated 661W cells.

**CONCLUSION**

We developed and validated a novel HPLC method that provides a rapid, simple, and accessible way to quantify the drug. This may facilitate improved understanding of the pharmacokinetic properties of GSK-J4, and provides a practical platform for testing novel formulations. This extraction protocol can also be adapted for isolating the drug from other cell cultures, or retinal tissue obtained after *ex vivo* or *in vivo* treatment, which is the focus of ongoing studies.